SMART INDIA HACKATHON 2025

TEAM DETAILS AND PROBLEM STATEMENT

- **Problem Statement ID- 25222**
- **Problem Statement Title- Digital Twin for CO_2 Capture in Urban Settings**
- **Theme- Clean & Green Technology**
- **PS Category- Software**
- **Team ID- 82627**
- **Team Name- ResolveX**

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IDEA & APPROACH DETAILS

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IDEA: A Complete software structure is a **Digital Twin Environment** that visualizes a city in 3D, mapping real-time or simulated CO_2 data from transport, industry, and households. It uses **dynamic emission and dispersion models** to simulate CO_2 patterns based on traffic, weather, and terrain. Users can test **interventions** like CO_2 capture units, green walls, or biofilters and compare their effectiveness. An **analytics dashboard** shows hotspots, reduction trends, and cost-benefit insights, while **AI-driven tools** support scenario comparison and location prioritization.

SOLUTION:

Modelling Emissions Across Transport, power and Industry:

❖ **Emission Estimation Framework:**
- Common principle: **Emissions = Activity × Emission Factor**
- Inputs include traffic data, fuel consumption, and production outputs
- Provides sector-wise emission inventory for analysis and visualization

❖ **Sectoral Modelling & Dispersion:**
- **Transport:** Based on **vehicle type**, **traffic flow**, and **speed** — highlights road-level hotspots
- **Power:** Uses generation or **fuel data**, accounts for **renewable vs. fossil mix**
- **Industry:** Includes **combustion** and **process emissions** from factories
- **Dispersion modeling:** Simulates how CO_2 spreads using weather and terrain data

❖ **Data Integration, Calibration & Interventions**
- Combines data from **sensors, utilities, and industries**
- **Calibrates** model outputs using real-time CO_2 sensor data for accuracy
- Simulates **carbon capture interventions** (e.g., green walls, roadside units)
- Evaluates impact on **air quality, emission reduction, and cost-effectiveness**
- Supports **decision-making** for sustainable urban planning

Carbon Capture Intervention:

- **Roadside Unit:** Point sink → **uses airflow & efficiency** → outputs CO_2 captured (kg/hr)
- **Vertical Garden:** Surface sink → **area & plant efficiency** → CO_2 captured (kg/m²/day)
- **Algae Bioreactor:** Volumetric sink → **growth rate & CO_2 level** → CO_2 captured (kg/day)
- **Tree Canopy:** Distributed sink → **tree count & growth stage** → CO_2 absorbed (kg/year)
- **Industrial CCS:** Emission modifier → **capture %** → net emission reduction (%)
- **MOF Panel:** Surface sink → **adsorption capacity** → CO_2 captured (kg/m²/day)

Carbon Capture Decision Support Dashboard

❖ **Overview and Twin:**
- Select **city/region** & **date/time**
- Show total **CO_2 capture**d, **emission reduction %, active vs planned units**
- **Point, surface, volumetric**, and **distributed layers** for each intervention
- **Heatmaps** for **CO_2 concentration**

❖ **Intervention Analysis:**
- Filter by intervention type (**technological, biological, hybrid, industrial**)
- Charts: **CO_2 captured over time, efficiency trends, type comparisons**

❖ **Simulation & Scenario Planning:**
- Adjust parameters: **airflow, area, growth rate, capture %, adsorption rate**
- Run **simulations & visualize results** in real time

❖ **Reporting & Exports**
- Summary tables and cumulative CO_2 capture
- **Alerts** for low-performing or over-capacity units

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TECHNICAL APPROACH

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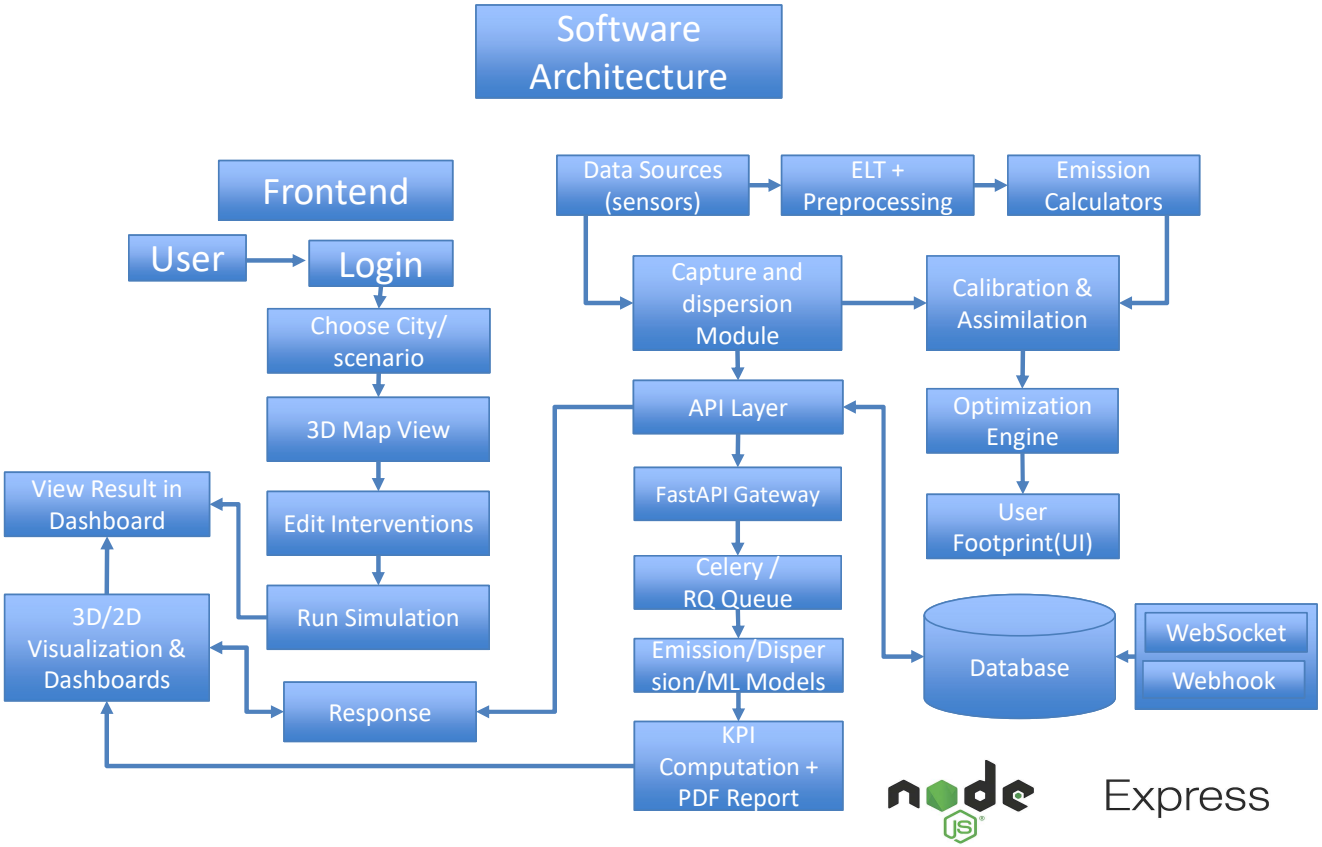

node JS Express

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Innovation and Uniqueness:

- **Integrated Multi-Sector Modeling**
  - Unifies **transport**, **power**, and **industrial emission modeling** in one **digital twin environment**, giving a complete picture of a city's carbon ecosystem.
- **Dynamic, Real-Time Simulation**
  - Uses **live traffic, energy, and weather data** to simulate **CO_2 generation and dispersion** over time, enabling **realistic and time-sensitive analysis**.
- **AI-Powered Predictive Insights**
  - Employs **machine learning** to forecast **future emission hotspots**, **optimal intervention areas**, and **long-term carbon reduction trends**.
- **Interactive 3D Decision Platform**
  - Offers **3D visualization**, **heatmaps**, and **scenario testing tools** that let users **virtually experiment** with interventions like **biofilters** or **green walls** before real-world deployment.
- **Scalable and Sustainable Design**
  - Built on a **modular architecture** that can integrate with **IoT sensors**, **smart city systems**, and **renewable energy models**, ensuring **future adaptability and sustainability**.

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FEASIBILITY AND VIABILITY

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Feasibility:

- **Technical Feasibility:**
  - 3D city modeling is achievable using open-source data (**OSM**, **CityJSON**) and libraries like **CesiumJS** or **Deck.gl**.
  - Simplified dispersion modeling (**grid-based or Gaussian puff**) is computationally feasible for **real-time** or **near-real-time simulation**.
  - **ML-based hotspot prediction** and scenario optimization are possible with modern frameworks (**PyTorch, scikit-learn**).
- **Data Feasibility:**
  - **Open-source geospatial** and **traffic data** are available for many cities.
  - Building energy use can be estimated using standard energy intensity metrics if real data is unavailable.
  - Weather data (**wind**, **temperature**) can be sourced from **APIs or historical datasets.**
- **Resource Feasibility:**
  - Moderate computational resources are sufficient for **baseline modelling**.
  - **Cloud computing** can support large-scale or high-fidelity simulations cost-effectively.
- **Operational Feasibility:**
  - Web-based platform enables accessibility for **planners**, **researchers**, and **decision-makers**.
  - Low maintenance requirement once data pipelines and visualization modules are established.

Viability

Economic Viability

Can be positioned as a decision-support tool for city planners, environmental agencies, and sustainability consultants

Cloud-based architecture,careduce uptront hardware costs

Social & Environmental Viability

Directly supports sustainability goals and emissions reductior commitments

Encourages public engagement and transparency in environmental management

Promotes cleaner, healthier urban living

Market Viability

Increasing demand for digitai-twin-based urban solat-lons worldwide

Can be adopted by municipalities, research institutions, and urban developers

Risk Viability

Main risks involve data quality, computational limitations, and user interpretation

These can be miti-gated by model calibration, modular scaling, and validation using sensor data

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IMPACT AND BENEFITS`

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Impacts:

❖ **Environmental Impact**

- Identifies **CO_2 hotspots** and supports **targeted emission reduction** strategies.
- Enables **continuous monitoring** of **urban air quality** and **carbon levels**.
- Encourages **green interventions** like **vertical gardens**, **biofilters**, and **capture units**.
- Promotes **data-backed climate resilience** and **sustainable infrastructure**.

❖ **Societal Impact**

- Increases **public awareness** of **air pollution** and its **health impacts**.
- Supports **government initiatives** for **sustainable urban development**.
- Leads to **improved public health** by reducing exposure to **harmful emissions**.
- Fosters **community participation** in **climate action**.

❖ **Technological Impact**

- Advances **digital twin technology** for **environmental modeling**.
- Integrates **real-time data**, **simulation**, and **AI-driven prediction**.
- Encourages **innovation** in **urban analytics**, **IoT**, and **smart infrastructure**.
- Strengthens **research and development** in **climate tech** and **smart cities.**

❖ **Policy Impact**

- Provides **scientific support** for **urban emission policies** and **green planning**.
- Helps authorities **evaluate policy effectiveness** through **scenario simulations**.

Benefits:

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RESEARCH AND REFERENCES

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